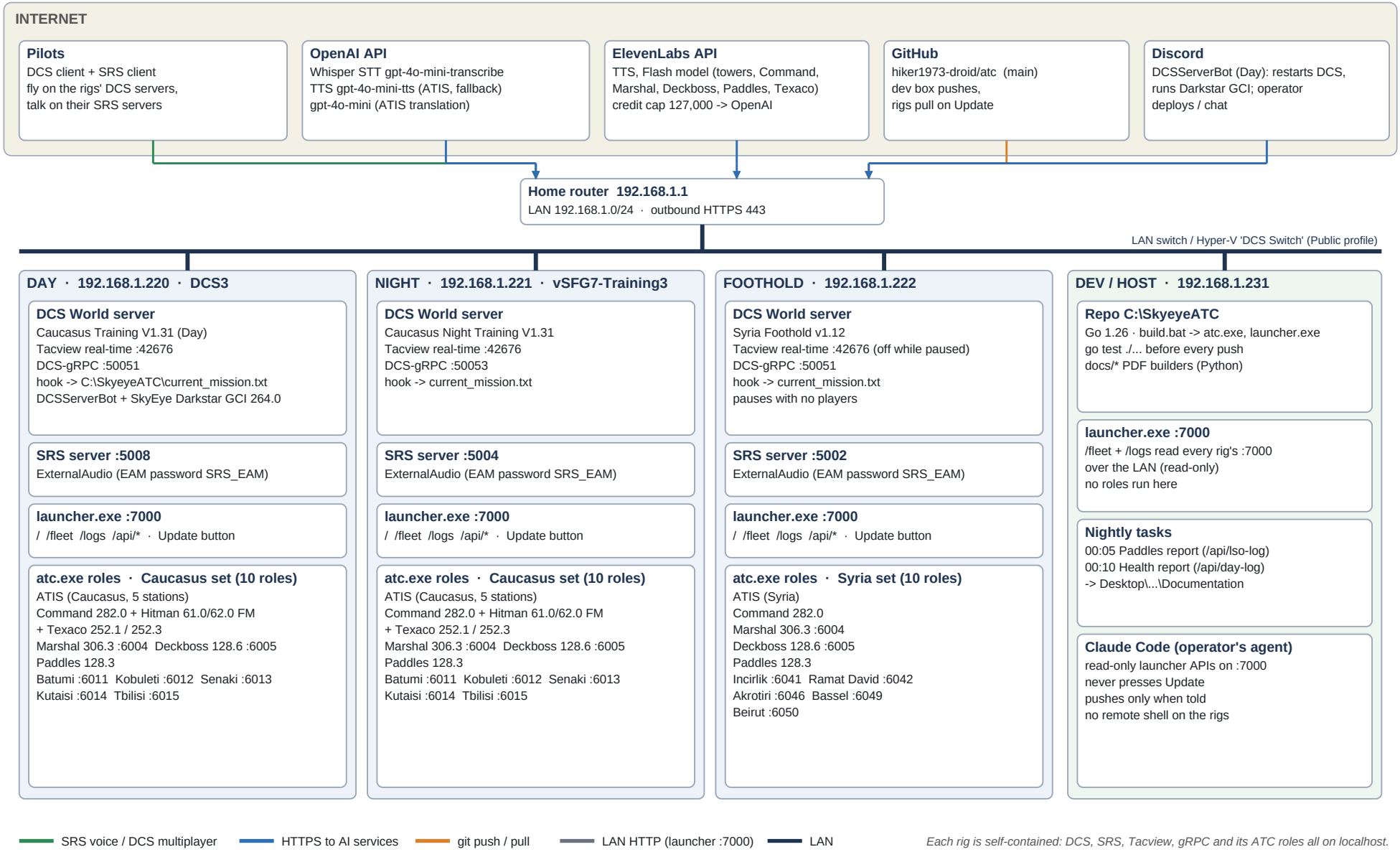
SkyEye ATC Network Architecture

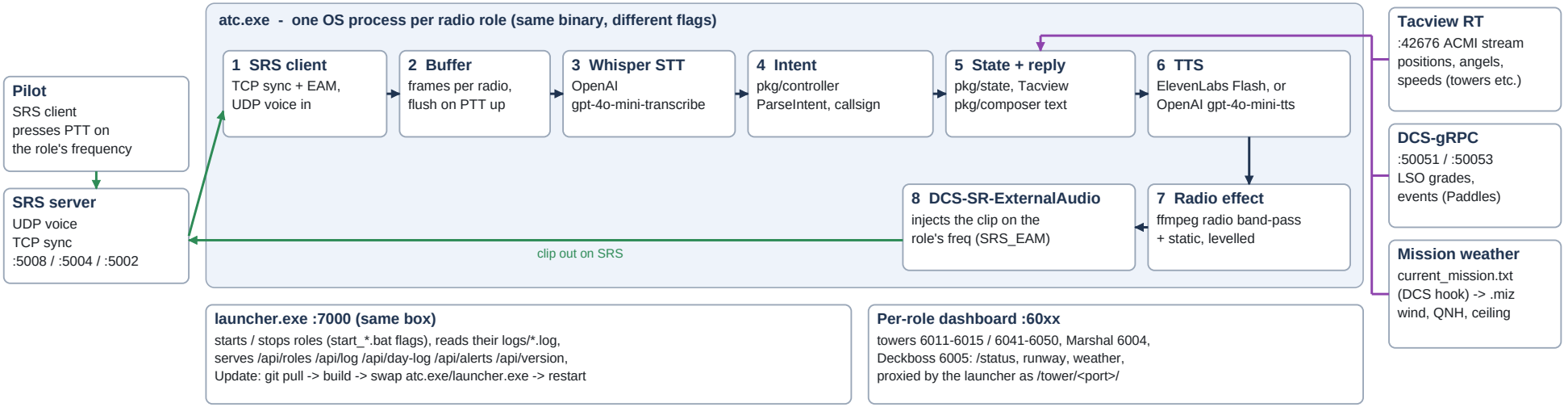
vSFG-7 · three DCS rigs and the dev box on one home LAN · SkyEye 7047a8a, 20 Oct 2026



Each rig is self-contained: DCS, SRS, Tacview, gRPC and its ATC roles all on localhost.

Inside a rig: one role's voice loop

Every radio role (each tower, ATIS, Command with Hitman and Texaco, Marshal, Deckboss, Paddles) is its own **atc.exe** process, started by a start_*.bat with that role's flags. All of them talk to the rig's own SRS, Tacview and DCS-gRPC on localhost, and to OpenAI / ElevenLabs over HTTPS.



Ports and services

Service	Port	Used by	Set by
SRS server	TCP+UDP 5008 (Day) · 5004 (Night) · 5002 (Foothold)	pilots (internet), every atc.exe role	**SKYEYE_SRS** = localhost:<port> on the rig
Tacview real-time	TCP 42676	towers, Marshal, Deckboss, Paddles, Command, launcher	**SKYEYE_TACVIEW**. DCS: Options > Special > Tacview > real-time telemetry
DCS-gRPC	TCP 50051 (Day, Foothold) · 50053 (Night)	Paddles (LSO grades, events)	start_lso_*.bat picks it; **SKYEYE_GRPC** overrides
Launcher	TCP 7000	LAN browsers, the dev box, the nightly reports	firewall rule must allow the **Public** profile on the Hyper-V switch
Role dashboards	TCP 6004-6005, 6011-6015, 6041-6050 (others per theatre 6021-6054)	the launcher (proxy /tower/<port>/), operators	--dashboard-port in each start_*.bat
pprof (Command)	TCP 7770, localhost only	debugging goroutines	--pprof-port
OpenAI API	HTTPS 443 out	every role (STT), ATIS + fallback TTS	**OPENAI_API_KEY**
ElevenLabs API	HTTPS 443 out	TTS for the ElevenLabs voices; credit reads every 10 min	**ELEVENLABS_API_KEY**; --elevenlabs-stop-at 127000
GitHub	HTTPS 443 out	Update (git pull --ff-only); dev box push	repo github.com/hiker1973-droid/atc

Builder's guide: stand up a new rig

From a bare Windows box to a rig on the fleet. Steps 1-4 are DCS and SRS, 5-10 are SkyEye.

Build

#	Step	What to do
1	Box and network	Windows 10/11 or Server, static IP on 192.168.1.0/24 (gateway 192.168.1.1). Note the hostname - two boxes already answer "DCS3", so always identify a rig by IP.
2	DCS World dedicated server	Install, load the mission set (Caucasus Training / Night Training V1.31, or Foothold Syria). Turn on **Tacview real-time telemetry on 42676**. For Paddles: DCS-gRPC enabled (MissionScripting.lua desanitised, gRPC on :50051, or :50053 on a second server on the same box).
3	Mission hook	Copy **tools/dcs-hooks/vsfg7-current-mission.lua** to *Saved Games\DCS.dcs_serverrelease\Scripts\Hooks*, restart DCS once. Each load writes C:\SkyeyeATC\current_mission.txt and re-pins SKYEYE_MIZ; every role follows the weather live.
4	SRS server	Install DCS-SimpleRadio Standalone, pick the port (5008 / 5004 / 5002 are taken by the current rigs), turn on the External Audio Module and set its password. ExternalAudio must be at *C:\Program Files\Eagle Dynamics\SRS\ExternalAudio\DCS-SR-ExternalAudio.exe* (or pass --external-audio).
5	Tools	Git, **Go 1.26** (winget install GoLang.Go), ffmpeg (default C:\ffmpeg-master-latest-win64-gpl\bin\ffmpeg.exe, or pass --ffmpeg - a wrong path fails quietly: clean audio, no radio effect), Python 3 only for the docs.
6	Code	**git clone https://github.com/hiker1973-droid/atc C:\SkyeyeATC** (every bat assumes this path), then **build.bat** -> atc.exe + launcher.exe. Build from a real clone, not a git worktree, or /api/version says "unknown".
7	Environment (setx, then a new window)	**OPENAI_API_KEY**, **ELEVENLABS_API_KEY**, **SRS_EAM** (EAM password), **SKYEYE_SRS** = localhost:<srs port>, **SKYEYE_TACVIEW** = localhost:42676, optional SKYEYE_GRPC. SKYEYE_MIZ is set by the hook. Run **check_env.bat**: all set, SRS and Tacview REACHABLE.
8	Firewall and AV	Inbound TCP 7000 (and the dashboard ports if wanted) on the LAN, for **Public** as well as Private. Add an antivirus exception for C:\SkyeyeATC - Bitdefender has quarantined launcher.exe before and then blocked writes to that path.
9	Start	**start_launcher.bat**, open http://<ip>:7000/, start the theatre (Start region), or run start_region_<theatre>.bat. Each role opens its own console window. Never start regions on the dev box (no SRS_EAM, bats pause).
10	Join the fleet	Add the rig to the host launcher's **--fleet** list (name@ip:7000) so /fleet, /logs and the nightly reports see it. Add it to rigcheck / run_health / run_nightly host lists.

Deploying a change (the Update button)

#	Step
1	Developer: tests pass, commit, **git push** to main (dev box).
2	Operator: /fleet **Update** (per rig) or Update all rigs.
3	Launcher: git pull --ff-only, build atc.new.exe + launcher.new.exe, re-read keys from the registry.
4	Stop running regions, swap atc.exe (old kept as atc.prev.exe), start the regions again (~30 s off air).
5	Swap launcher.exe, restart it; result in update_last.json. Verify with the checklist below.

Verify a rig (read-only, from any LAN browser: http://<ip>:7000)

Check	Expect
/api/version	build commit (e.g. 7047a8a) - the BUILD, not what was pulled; modified=true is normal after Update
/api/roles	every role of the region status=running (Caucasus: 10)
/api/log?name=<role>	"ATC online" / "Command channel starting"; tankers: "Tanker net starting"; "Weather loaded from .miz" naming the loaded mission
/api/miz-weather	mizName matches the mission DCS is running
/api/alerts	null, or only known items (Foothold paused: Tacview unreachable)
On the radio	radio check on a tower and on Command: loud and clear in the expected voice

Known faults

Symptom	Cause / fix
TACVIEW OFFLINE on every role	Real-time telemetry off in DCS, or the mission is paused (Foothold with no players - normal).
SRS TCP failed for ~1 min, then fine	Mission rotation restarting DCS/SRS. Roles re-register on their own.
Dashboard :7000 dead, ATC still talking	launcher.exe died; start_launcher.bat (roles keep running).
Radio effect missing (clean audio)	ffmpeg path wrong - pass --ffmpeg.
Voices switched to OpenAI	ElevenLabs credit cap (127,000) reached; resets on the 14th, 16:31Z.
Roles on the wrong weather	Hook not installed / current_mission.txt stale; check /api/miz-weather.
ATIS ExternalAudio crashes, 45 s TTS timeouts	Known, left alone (ruling 2026-10-19); ATIS resends.

No remote shell reaches the rigs (WinRM / SSH off): pulls, builds and restarts happen through the Update button or at the box. The dev box only reads the launchers.

Document builders in the repo

Document	Builder	Run	Notes
Comms Packet (16:9 PDF, 26 pages)	docs/comms-packet/_tools/build_skyeye_packet.py	python "_tools\build_skyeye_packet.py" (from docs/comms-packet)	VER / DATE / build stamp; NOTES.md entry
Phraseology Reference	docs/phraseology/build_phraseology.py	python -I docs/phraseology/build_phraseology.py	facts.json / phrases.json; drift tests TestDocFacts / TestDocPhrases
Texaco Tanker Comms Flow	docs/tanker-flow/build_tanker_flow.py	python -I docs/tanker-flow/build_tanker_flow.py	one page; copies to Documentation
Paddles LSO Comms Flow	docs/paddles-flow/build_paddles_flow.py	python -I docs/paddles-flow/build_paddles_flow.py	
Hitman JTAC Comms Flow	docs/hitman-flow/build_hitman_flow.py	python -I docs/hitman-flow/build_hitman_flow.py	
Paddles daily report	docs/paddles-report/run_nightly.py -> build_paddles_report.py	task "SkyEye Paddles Report" 00:05	reads /api/lso-log from each rig
Health daily report	docs/health-report/run_health.py -> build_health_report.py	task "SkyEye Health Report" 00:10	reads /api/day-log, /api/version, ElevenLabs usage
This document	docs/network-architecture/build_network_architecture.py	python -I docs/network-architecture/build_network_architecture.py	edit the tables when a rig changes